

Andrei Dobrete

Curriculum Vitae

in [linkedin.com/in/Andy3153](https://www.linkedin.com/in/Andy3153)

 github.com/Andy3153

 gitlab.com/Andy3153



Self-taught Systems Engineer and Robotics Engineering Graduate focused on network/server management, Linux, NixOS, Docker and shell/Python automation. Hands-on experience in industrial PLC automation and web development.

Work Experience

07/2025 – 08/2025

Automation & Electrical Engineering Intern

IMSAT S.A. | imsat.ro

- Implemented electrical and control-system modifications at the Ford Otosan plant in Craiova, commissioning and troubleshooting conveyor systems used for door-accessory assembly.
- Programmed Allen-Bradley PLCs using software like RSLogix 5000 and developed HMI applications (FactoryTalk View Studio for PanelView Plus 1000), also used Connected Components Workbench for supporting controllers.
- Created and revised 3D CAD models (Siemens NX, CATIA V5) to validate component fit and support mechanical-controls integration.

06/2024 – 08/2024

Full-Stack Web Development Intern

EVASTAFF | evastaff.ro

- Implemented and maintained WordPress and Shopify e-commerce platforms on dedicated hosting using cPanel, including server configuration and deployment.
- Designed and developed complete online stores: storefronts, product listings, shopping cart and checkout flows.
- Automated sales and order-fulfillment processes to streamline the selling lifecycle.
- Performed front- and back-end customizations (PHP, CSS, SQL) and developed custom themes/plugins.
- Integrated third-party services and payment gateways via APIs to connect inventory, shipping and CRM systems.

Education and Training

03/2025 – Present

Cisco Certified Network Associate (CCNA)

Hackademy | hackademy7.wordpress.com

- Completed Semesters 1 & 2, covering Network Fundamentals, Switching, Routing, and Wireless Essentials.

10/2022 – 06/2026

Bachelor of Engineering (B.Eng.) in Robotics

Universitatea Națională de Știință și Tehnologie Politehnica București – Faculty of Industrial Engineering and Robotics | fiir.upb.ro

- **Thesis:** "Robotic palletizing cell with two inputs and two outputs integrating an articulated arm industrial robot equipped with a multifunctional effector"

Publications

05/2023

Phone Controlled Car | github.com/Andy3153/phone-controlled-car

Submissions:

- Student Scientific Communications Session of the Faculty of Industrial Engineering and Robotics, 2023 (industrialengineeringjournal.eu/pdf/vol7_2023_4/Art9.pdf)
- Journal of Industrial Engineering and Robotics, Volume 7, Number 4, 2023 (industrialengineeringjournal.eu/pdf/vol7_2023_4/vol7_2023_4.pdf, page 54)

Projects

Present

Systems Engineering & Self-Hosted Infrastructure

- Design and manage NixOS configurations for multi-machine environments (my personal computers and my home server), ensuring consistency and declarative system management.
- Author and maintain custom Nix derivations and reproducible development environments.
- Build and operate Docker Compose deployments and container images for self-hosted server deployments.
- Implement shell and Python automation scripts to streamline daily system operations and maintenance tasks.
- Curate and version-control application configurations (dotfiles) and $\text{\LaTeX}$ templates for portability and reproducibility.
- Contribute upstream through pull requests, maintaining forks, and submitting patches.

10/2019 – 10/2022

Lead Programmer, BroBots (FIRST Tech Challenge RO142 | 19088)

- Primary developer for the robot control codebase using Java, responsible for implementation, debugging and version control.
- Heavily contributed to mechanical design, assembly and prototyping.
- Contributed to iterative testing and performance tuning for competition readiness.
- Mentored newer team members and facilitated cross-functional collaboration between software and mechanical subteams.

Digital Skills

Infrastructure	○ Linux	○ Docker/Podman	○ System Administration
	○ NixOS	○ systemd	○ QEMU/KVM
Development	○ Python	○ Lua	○ Java
	○ Nix	○ C	○ Ladder Logic
	○ POSIX Shell (bash/sh)	○ C++	
CAD/CAM	○ Git	○ Arduino	
	○ Siemens NX	○ AutoCAD	○ FreeCAD
Office	○ CATIA V5	○ Blender	
	○ $\text{\LaTeX}$	○ Microsoft Office	○ LibreOffice

Volunteering

03/2024 – Present

Volunteers Polytechnic Bucharest – Member | unstpb.ro

10/2022 – 06/2023

The Student Organization of the Faculty of Industrial Engineering and Robotics – Member | osfiir.ro

07/2019 – 12/2022

National Red Cross Society Râmnicu Vâlcea – Member | crucearosievalcea.ro

Language Skills

Romanian

Native

English

B2 (Professional Working Proficiency)

Driving License

08/2021

Full EU – Categories **B**, **B1**, **AM**